ON THE STRUCTURE AND ELECTRONIC SPECTRA OF EXCITED NEUTRAL ARGON RARE GAS CLUSTERS USING HOLE PARTICLE PSEUDOPOTENTIAL METHOD

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The structures of neutral excited Argon clusters are investigated using Hole-Particle Pseudopotential (HPP) [1] and compared with previous results [2] obtained using Diatomic-In-Molecules (DIM) method parametrized using CI potential energy curves [3].

The global minimum structures for excited Argon for $N \leq 13$ are explored which differ from what was observed previously using diatomic-in-molecules approach. To parametrize DIM, the effect of diabatization is also investigated. The lowest energy isomer of trimer is observed to be linear anti-symmetric (C2v symmetry) instead of linear-symmetric (D∞h symmetry) as shown in figure 1.

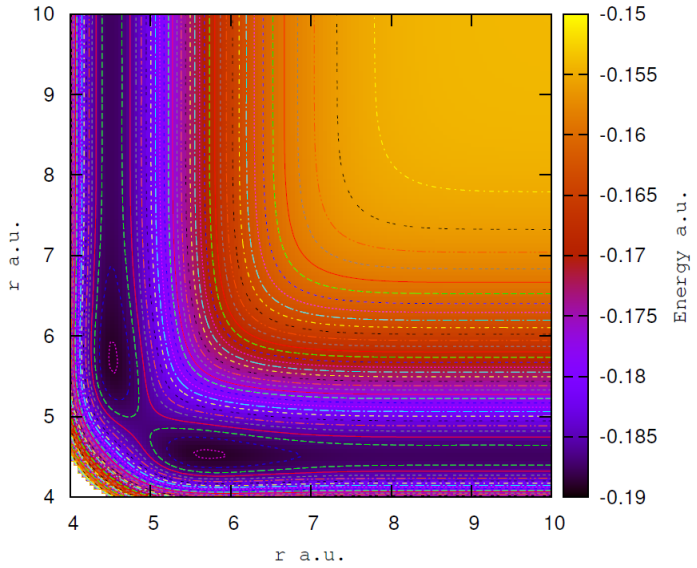


Figure 1: Potential Energy surface for excited Argon trimer linear system.

References:

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[2] Naumkin, F. Y., & Wales, D. J. (1999). Rydberg excitations in rare gas clusters: structure and electronic spectra of Ar* n (3≤ n≤ 25). Molecular Physics, 96(9), 1295-1304.

[3] Spiegelmann, F., & Gadea, F. X. (1984). Calcul théorique en couplage Λ-Σ du spectre électronique des excimères Ar2 et Kr2 corrélé aux configurations atomiques np5 (n+ 1) s et np5 (n+ 1) p. Journal de Physique, 45(6), 1003-1023.

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